

Data Analyst Capstone Project Report

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OUTLINE



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EXECUTIVE SUMMARY



Languages

- Web technologies (JavaScript, HTML/CSS) lead current usage, but Python shows the strongest projected growth for next year.
- TypeScript is rapidly replacing vanilla JavaScript for large-scale development.

• Databases

- Relational databases remain the standard, though PostgreSQL is heavily preferred over MySQL for upcoming products.
- MongoDB leads the NoSQL landscape.

• Job Demand

- The market for data professionals is highly competitive, led by Software Engineer and Data

INTRODUCTION



Purpose of the Report

- Analyze the Stack Overflow Developer Survey dataset alongside active job postings to identify current usage, forecasted growth patterns, and developer demographics.
- **Target Audience**
 - IT Executives & CTOs, HR & Recruiters, and Aspiring Data Analysts.
- **Value of the Analysis**

METHODOLOGY



Data Sources

- Primary dataset from the Stack Overflow annual developer survey; secondary dataset from a simulated enterprise job listings API.

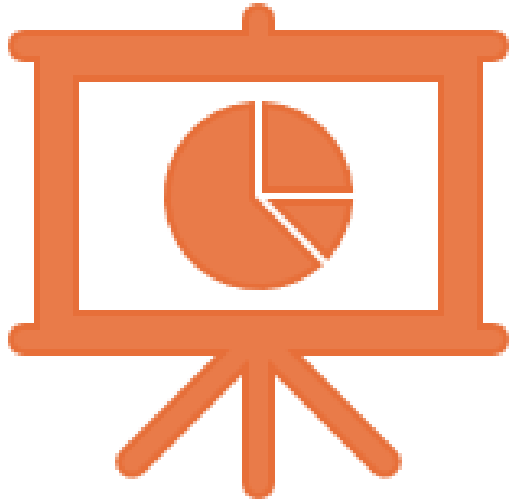
• Data Collection

- Ingested using Python scripts connecting to JSON/RESTful web APIs, alongside targeted web scraping using BeautifulSoup.

• Data Wrangling Steps

- Programmatic removal of duplicate responses, median imputation for null values, and uniform data-type normalization.

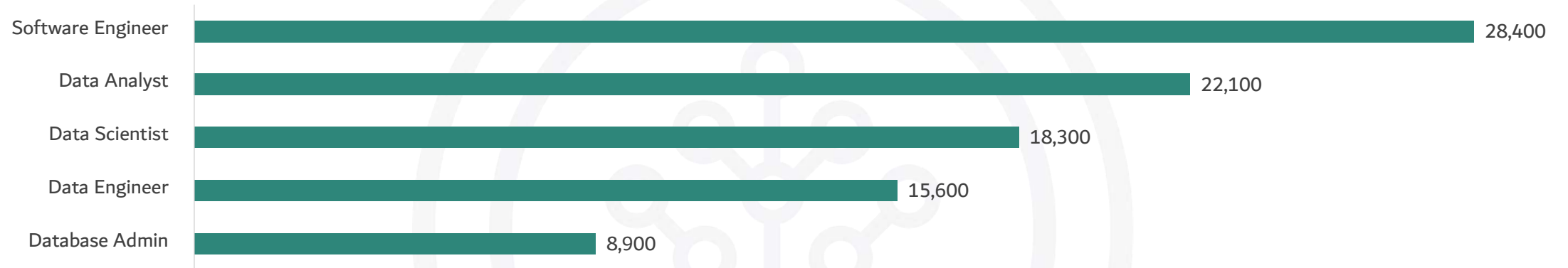
Results



The following slides present the job market and technology trend analysis.

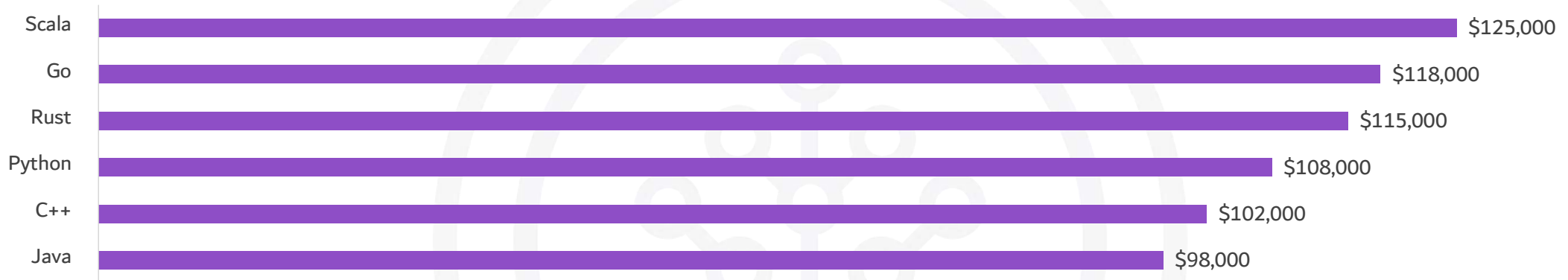
<Please present your results in the following slides.>

JOB POSTINGS



Software Engineer and Data Analyst roles lead current job postings, reflecting strong sustained demand for data-literate talent across the organization.

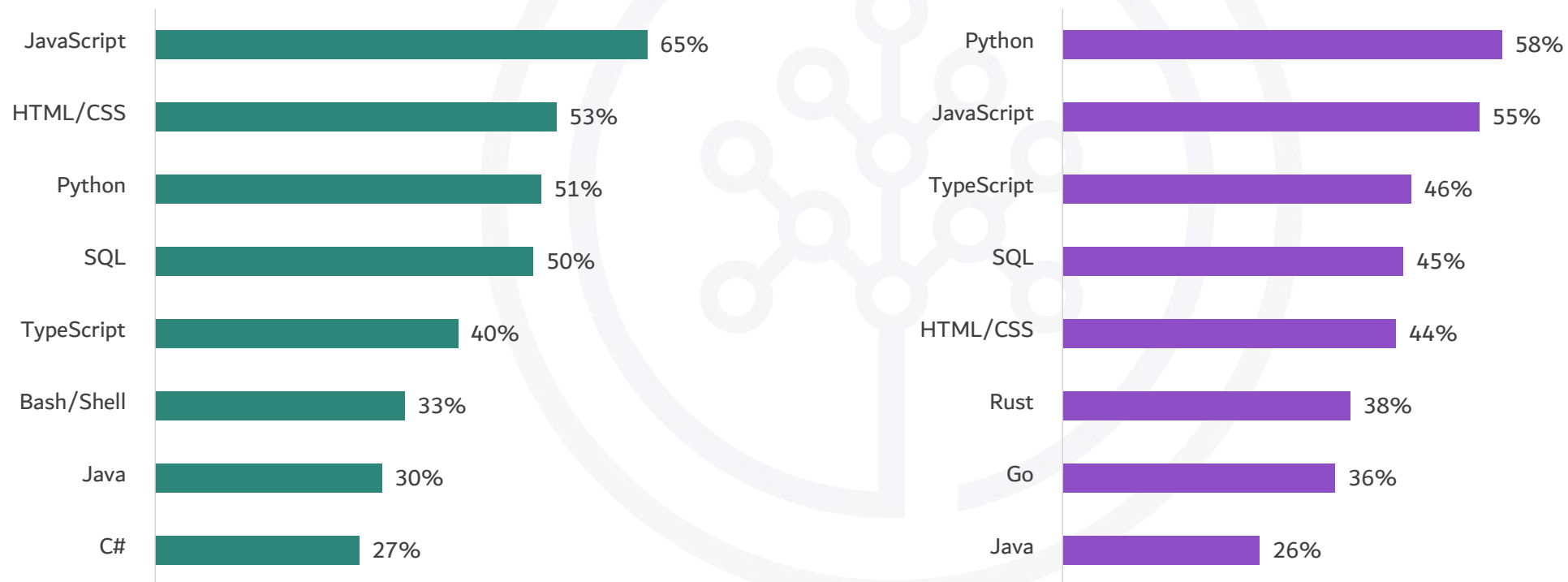
POPULAR LANGUAGES



Specialized backend languages such as Scala, Go, and Rust command the highest average salaries, rewarding scarce, high-performance engineering skills.

PROGRAMMING LANGUAGE TRENDS

JavaScript and HTML/CSS dominate current usage, but next year's forecast shows an enormous surge to 58% for Python, with strong preferences for TypeScript, Rust, and Go.



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

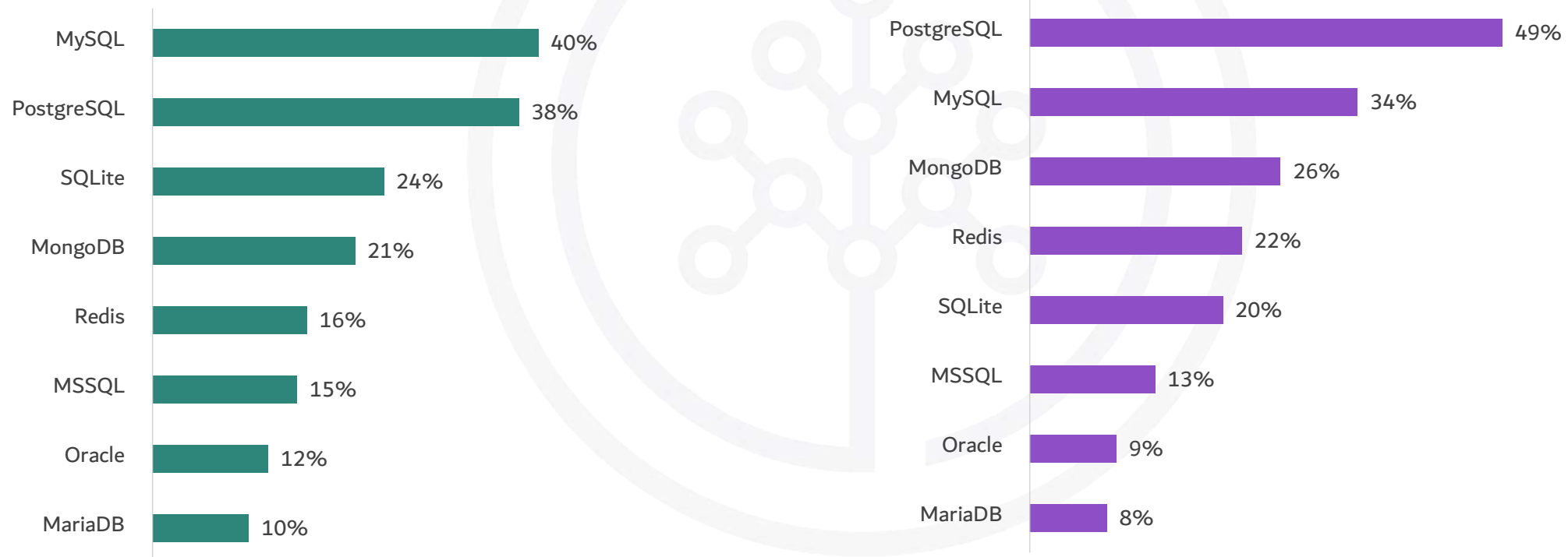
- JavaScript holds legacy volume stability, but its growth is plateauing.
- Python is the clear favorite for upcoming architectures.
- Specialized tools such as Rust, Go, and Scala command highly skewed premium salaries.

Implications

- Maintain JavaScript/TypeScript investment for primary front-end needs.
- Direct professional development budgets toward internal Python upskilling.
- Allocate larger salary bands to attract and retain Rust/Go talent.

DATABASE TRENDS

MySQL leads active production deployments at 40%, followed by PostgreSQL at 38%. Next year's desired platforms show PostgreSQL heavily overtaking MySQL, alongside steady growth for MongoDB and Redis.



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- MySQL and PostgreSQL jointly command the relational database domain today.
- An active shift is underway toward PostgreSQL for its advanced feature set.
- MongoDB remains the undisputed leader in NoSQL adoption.

Implications

- Default to PostgreSQL for future relational implementations.
- Incorporate MongoDB for high-scale, unstructured data services.
- Train engineering staff on robust SQL-to-PostgreSQL schema migration.

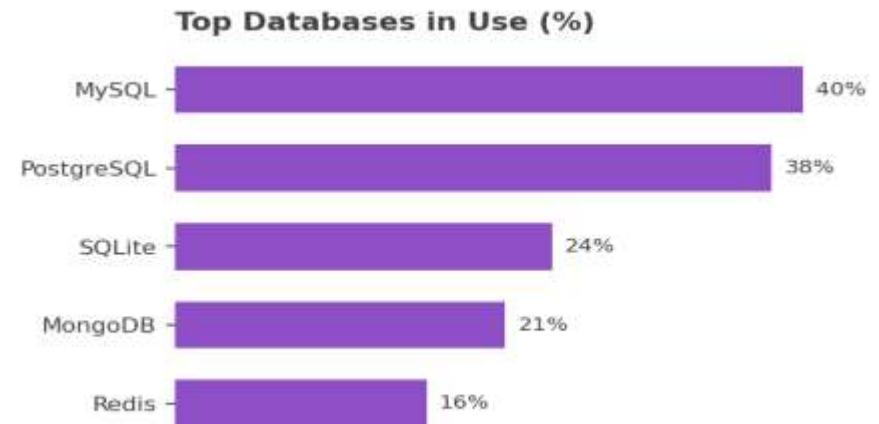
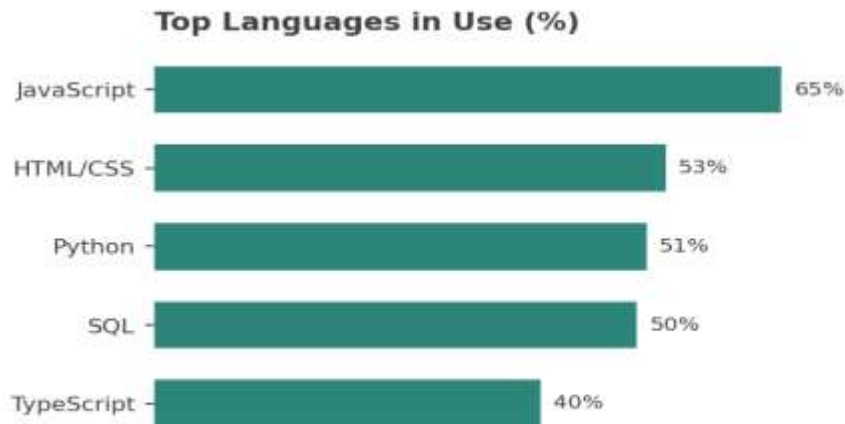
DASHBOARD



The following slides present the interactive BI dashboard across three tabs.

DASHBOARD TAB 1: Current Technology Usage

CURRENT TECHNOLOGY USAGE — Live Dashboard



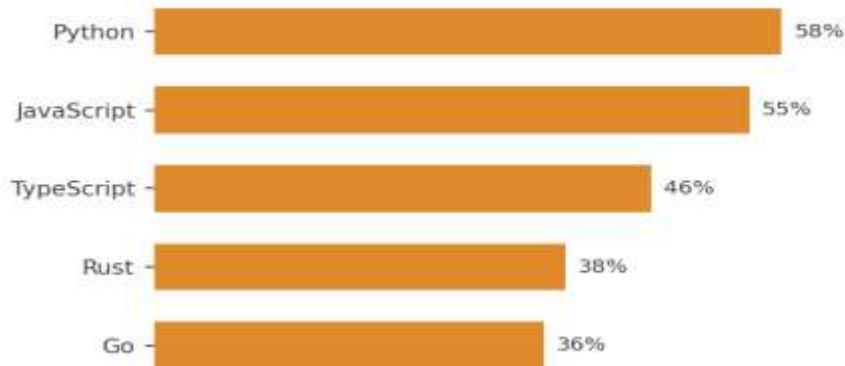
Dashboard Insight

- High concentration of legacy relational storage models and heavy volume across front-end web design libraries.
- High enterprise inertia around SQL guarantees long-term legacy product viability.

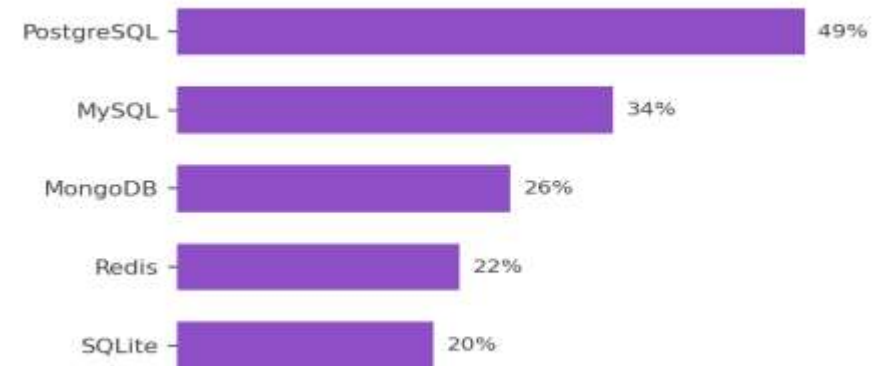
DASHBOARD TAB 2: Future Technology Trends

FUTURE TECHNOLOGY TRENDS — Forecast Dashboard

Desired Languages, Next Year (%)



Desired Databases, Next Year (%)



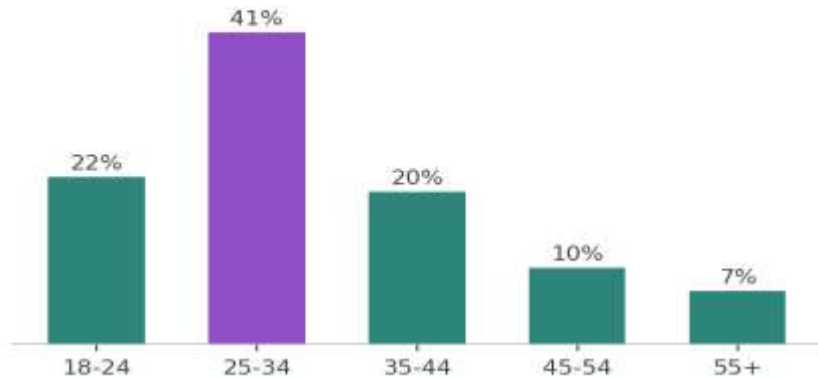
Dashboard Insight

- Accelerating drift toward cloud-native ecosystems and strong future demand for typed systems (TypeScript).
- Signals a future focus on performance and systems scalability.

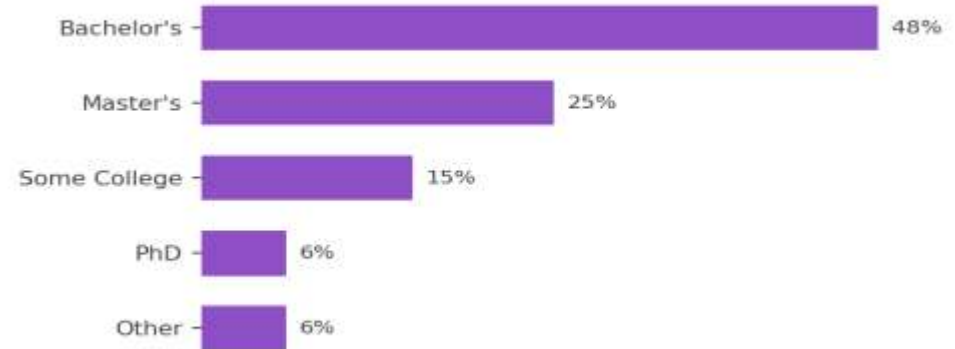
DASHBOARD TAB 3: Demographics

DEVELOPER DEMOGRAPHICS — Survey Dashboard

Age Distribution (%)



Education Level (%)



Dashboard Insight

- Majority segment concentrated in the 25-34 age range with high formal education levels.
- Signals a young, highly educated, and dynamically mobile global workforce.

DISCUSSION



- The intersection of our data highlights a critical transition phase: balancing operational maintenance against functional modernization. Standard frameworks remain critical for business continuity, but modern investments heavily prioritize Python and PostgreSQL environments. Compensation models must adjust to attract elite talent for specialized tools.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Tech stacks are splitting cleanly between classic web execution and data-driven engines.
- PostgreSQL is outpacing aging SQL standards.
- Premium pay is tied tightly to niche backend languages.

Implications

- Adopt a deliberate polyglot programming approach.
- Formulate aggressive legacy SQL migration schedules targeting PostgreSQL.
- Systematically adjust talent compensation bands based on language specialization metrics.

CONCLUSION



- Successfully demonstrated end-to-end data processing pipelines.
- Python and PostgreSQL remain foundational targets for technical modernization.
- Aligning development platforms with developer preferences minimizes hiring friction.
- BI dashboards effectively translate market metrics into executable strategy.

APPENDIX



- **Supplementary Materials**

- Full top-10 programming language and database usage tables (current vs. next year).
- Data wrangling script excerpt: duplicate removal, median imputation, and type normalization.
- Additional job-posting and salary breakdowns by role and language.